AKSHIT KUMAR

Updated: November 2026

Department of Management, University of Toronto Scarborough
Rotman School of Management (cross-appointed)
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RESEARCH INTERESTS

I am broadly interested in the operations of online platforms and marketplaces, with a particular focus on AI agents and recommendation systems.

ACADEMIC APPOINTMENTS

July 2026 - Present — **Assistant Professor, Operations Management**
Department of Management, University of Toronto Scarborough
Rotman School of Management (cross-appointed)

July 2025 - June 2026 — **Postdoctoral Researcher**
Center for Algorithms, Data and Market Design, Yale University

EDUCATION

Sep 2020 - May 2025 — **Ph.D., Operations Management**
Columbia University, Graduate School of Business
Advisors: Prof. Omar Besbes and Prof. Yash Kanoria

Jul 2018 - May 2020 — **M.S., Electrical and Computer Engineering**
University of Michigan
Advisor: Prof. Vijay Subramanian

Aug 2014 - May 2018 — **B.Tech., Electrical Engineering**
Indian Institute of Technology Madras
Advisor: Prof. Rahul Vaze

PUBLICATIONS

Working Papers

1. A. Kumar. Feature-Based Dynamic Matching in the Dark. [Link]
2. A. Kumar, V. Manshadi, A. Tumu. How Much Should a Conversational Recommender System Converse? [Link]
3. J. Anunrojwong, A. Kumar, R. Kumar. Trading with the STARS: Algorithm Design & Spectrum of Fundamental Limits for Trading with Storage. [Link]
4. O. Besbes, Y. Kanoria, A. Kumar. Impact of Rankings and Personalized Recommendations in Marketplaces. [Link]

Journal Papers

1. Y. Chen, Y. Kanoria, A. Kumar, W. Zhang. Feature-Based Dynamic Matching. *Operations Research*, 74(2):788-803, 2026. [Link]
   - Third Place, 2025 Michael H. Rothkopf Junior Researcher Paper Prize.
   - First Place, 2024 Jeff McGill RMP Best Student Paper Prize.
2. O. Besbes, Y. Kanoria, A. Kumar. Dynamic Resource Allocation: Algorithmic Design Principles and Spectrum of Achievable Performances. *Operations Research*, 73(3):1273-1288, 2025. [Link]
   - Second Place, 2024 Michael H. Rothkopf Junior Researcher Paper Prize.
   - Finalist, 2023 INFORMS George Nicholson Student Paper Competition.
   - Finalist, 2023 Jeff McGill RMP Best Student Paper Prize.

Peer Reviewed Conference Proceedings

1. A. Kumar. Feature-Based Dynamic Matching in the Dark. *WINE 2026: Conference on Web and Internet Economics*. [Link]
2. A. Kumar, V. Manshadi, A. Tumu. How Much Should a Conversational Recommender System Converse? *EC'26: ACM Conference on Economics and Computation*. [Link]
3. A. Allouah, O. Besbes, J. Figueroa, Y. Kanoria, A. Kumar. What Is Your AI Agent Buying? Evaluation, Biases, Model Dependence, & Emerging Implications of Agentic E-Commerce. *WWW'26: ACM Web Conference*. [Link]
4. O. Besbes, Y. Kanoria, A. Kumar. Impact of Rankings and Personalized Recommendations in Marketplaces. *EC'25: ACM Conference on Economics and Computation*. [Link]
5. A. Kumar, T. Peng, Y. Wu, A. Zeevi. Performance of LLMs on Stochastic Modeling Operations Research Problems: From Theory to Practice. *WSC'25: Proceedings of the 2025 Winter Simulation Conference*. [Link]
6. O. Besbes, Y. Kanoria, A. Kumar. The Fault in Our Recommendations: On the Perils of Optimizing the Measurable. *RecSys'24: ACM Conference on Recommender Systems*. [Link]
7. A. Kumar, P. Hegde, R. Vaze, A. Alloum, C. Adjih. Breaking the Unit Throughput Barrier in Random Access Protocol Based Distributed Systems. *NCC'23: 29th National Conference on Communications*. [Link]
8. Y. Chen, Y. Kanoria, A. Kumar, W. Zhang. Feature-Based Dynamic Matching. *EC'23: ACM Conference on Economics and Computation*. [Link]
9. O. Besbes, Y. Kanoria, A. Kumar. The Multi-secretary Problem with Many Types. *EC'22: ACM Conference on Economics and Computation*. [Link]
10. A. Jain, Z. Kapetanovic, A. Kumar, V. N. Swamy, R. Patil, D. Vasisht, R. Sharma, M. Swaminathan, R. Chandra, A. Badam, G. Ranade, S. Sinha, A. U. Nambi. Low-cost aerial imaging for small holder farmers. *COMPASS'19: Proceedings of the 2nd ACM SIGCAS Conference on Computing and Sustainable Societies*. [Link]
    - Best Paper Award at COMPASS'19.
11. P. Hegde, A. Kumar, R. Vaze. Speed scaling under QoS constraints with finite buffer. *WiOpt'18: 16th International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks*. [Link]

AWARDS

2026	Finalist, JFIG Junior Faculty Interest Group Paper Competition
2025	Third Place, Michael H. Rothkopf Junior Researcher Paper Prize
2024	First Place, Jeff McGill RMP Best Student Paper Prize
2024	Second Place, Michael H. Rothkopf Junior Researcher Paper Prize
2024	Rising Star, Stanford Management Science and Engineering
2023	Finalist, INFORMS George Nicholson Student Paper Competition
2023	Finalist, Jeff McGill RMP Best Student Paper Prize
2023 - 2024	Deming Doctoral Fellowship, Columbia Business School
2023	Narula Doctoral Fellowship, Columbia Business School
2019	Best Paper Award, COMPASS'19
2016	Industry Category Winner at Microsoft Global Hackathon
2014	Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship
2011	National Talent Search Examination (NTSE) scholarship

PATENTS

1. R. Chandra, M. Swaminathan, V. N. Swamy, Z. Kapetanovic, D. Vasisht, A. Kumar, A. Badam, G. Ranade, S. Sinha, R. Patil. Low-cost, Long-term Aerial Imagery. [Link]
2. R. Chandra, M. Swaminathan, V. N. Swamy, Z. Kapetanovic, D. Vasisht, A. Kumar, A. Mehra, A. Gupta, S. Sinha, R. Patil. Aerial Imaging of a Region Using Above Ground Aerial Camera Platform. [Link]

INDUSTRY INTERNSHIPS

Jun 2023 - Sep 2023	**Amazon SCOT**, Bellevue, WA Developed multi-objective optimization algorithms for order fulfillment.
May 2018 - Aug 2018	**Nokia Bell Labs**, Paris, France Developed decoding schemes for distributed wireless systems.
Jun 2016 - Aug 2016	**Microsoft Research**, Bangalore, India Built low-cost precision-agriculture tools for smallholder farmers.

PROFESSIONAL SERVICE

- **Journal Reviewer:** Mathematics of Operations Research, Operations Research, Management Science, Manufacturing and Service Operations Management, Marketing Science, Information Systems Research
- **Program Committee Member:** EC 2026
- **Conference Reviewer/Sub-Reviewer:** MSOM Main Conference (2025, 2026)
- **Workshop Reviewer:** NeurIPS Workshop on Machine Learning and Operations Research (MLxOR), 2026
- Co-organizer for the 2024 NYC Operations Day PhD Colloquium.
- DRO Internal Seminar series co-organizer (2022-2024).
- DRO PhD student representative (2022-2024).

PRESENTATIONS

What Is Your AI Agent Buying? Evaluation, Biases, Model Dependence, and Emerging Implications for Agentic E-Commerce

- INFORMS Annual Meeting, San Francisco, CA — Nov 2026
- Workshop on Unstructured Data and Language Models, Ann Arbor, MI — Jul 2026
- ACM Web Conference (Poster), Online — Jul 2026
- ESIF Economics and AI+ML Meeting, Ithaca, NY — Jun 2026
- Behavioral Operations Conference, Tempe, AZ — May 2026
- Business Implications of Generative AI (Big.AI MIT), Boston, MA — Apr 2026
- Workshop on Information Systems and Economics (WISE), Nashville, TN — Dec 2025
- MIT Conference on Digital Experimentation (CODE), Boston, MA — Nov 2025
- Consumer Reports, New York, NY — Nov 2025
- Rotman Young Scholar Seminar, Online — Nov 2025
- Yale Computer Science and Economics Day, New Haven, CT — Sep 2025
- Workshop on AI and the Future of Collaborative Innovation, New York, NY — Sep 2025

Feature-Based Dynamic Matching in the Dark

- INFORMS Annual Meeting, San Francisco, CA — Nov 2026

How Much Should a Conversational Recommender System Converse?

- INFORMS Annual Meeting, San Francisco, CA — Nov 2026

Impact of Rankings and Personalized Recommendations in Marketplaces

- INFORMS Annual Meeting, Atlanta, GA — Oct 2025
- Revenue Management and Pricing (RMP) Conference, New York, NY — Jul 2025
- EC'25: ACM Conference on Economics and Computation, Stanford, CA — Jul 2025

Dynamic Resource Allocation: Theory and Practice

- From Matching to Markets Workshop, Corsica, France — May 2025
- Industrial and Systems Engineering, University of Minnesota, Minneapolis, MN — Jan 2025
- Mendoza College of Business, University of Notre Dame, Notre Dame, IN — Jan 2025
- Yale School of Management, New Haven, CT — Jan 2025
- Stanford Graduate School of Business, Stanford, CA — Jan 2025
- Cornell ORIE, Ithaca, NY — Dec 2024
- Cornell Tech ORIE, New York, NY — Dec 2024
- Imperial College Business School, London, England — Dec 2024
- Rotman School of Management, University of Toronto, Toronto, Canada — Dec 2024
- Indian School of Business, Hyderabad, India — Nov 2024
- NUS Business School, Singapore — Nov 2024
- NYU Stern School of Business, New York, NY — Nov 2024
- Wharton School of the University of Pennsylvania, Philadelphia, PA — Nov 2024

The Fault in Our Recommendations: On the Perils of Optimizing the Measurable

- Yale FDS Workshop on Social Algorithms (Poster) — Oct 2025
- Netflix ML Seminar, Online — Nov 2024
- RecSys'24: ACM Conference on Recommender Systems, Bari, Italy — Oct 2024

Dynamic Resource Allocation: Algorithmic Design Principles and Spectrum of Achievable Performances

- Workshop on Operations Research and Data Science (WORDS), Duke Fuqua, Durham, NC — Nov 2024
- Operations Rookiepalooza, Kellogg School of Management, Chicago, IL — Oct 2024
- INFORMS Annual Meeting, Seattle, WA — Oct 2024

Cornell Young Research Workshop, Ithaca, NY — Oct 2024
Purdue Operations Conference, West Lafayette, IN — Aug 2024
Industrial Engineering and Operations Research (IEOR) Seminar, IIT Bombay, Mumbai, India — Aug 2024
International Symposium on Mathematical Programming (ISMP), Montreal, Canada — Jul 2024
Revenue Management and Pricing (RMP) Conference, Los Angeles, CA — Jul 2024
MS&E Rising Stars Workshop, Stanford, CA — Apr 2024
INFORMS Annual Meeting, Phoenix, AZ — Oct 2023
Fulfillment Optimization Research Series, Amazon — Aug 2023
Tata Institute of Fundamental Research (TIFR), Mumbai, India — May 2023

Feature-Based Dynamic Matching

INFORMS Annual Meeting, Atlanta, GA — Oct 2025
INFORMS Annual Meeting, Seattle, WA — Oct 2024
Manufacturing and Service Operations Management (MSOM) Conference, Minneapolis, MN — Jul 2024
Marketplace Innovation Workshop, Online — May 2024
Revenue Management and Pricing (RMP) Conference, London, England — Jun 2023

A PROOF: Approximately PaReto Optimal Order Fulfillment

Deming Doctoral Fellowship Seminar, Columbia Business School — May 2024

The multi-secretary problem with many types

INFORMS Annual Meeting, Indianapolis, IN — Oct 2022
EC'22: ACM Conference on Economics and Computation, Boulder, CO — Jul 2022
Manufacturing and Service Operations Management (MSOM) Conference, Munich, Germany — Jun 2022
Revenue Management and Pricing (RMP) Conference, Online — Jun 2022

SKILLS

Programming: Python, C/C++, MATLAB, HTML/CSS
Tools: Git, LaTeX